



Exploring Product Relationships in E-commerce: A Data-Driven Approach Using the Apriori Algorithm

ABSTRACT

Mining an online retail transactional dataset with the Apriori algorithm to uncover frequent item sets and association rules, translating hidden purchasing patterns into actionable guidance for bundling, placement, and targeted promotions.

Business Questions

BQ1 · CO-PURCHASE

Which products are most frequently purchased together, and how can we leverage this for bundling, cross-selling, and promotions?

BQ2 · PLACEMENT

How can we optimize physical store layouts and e-commerce navigation to improve customer experience and grow average basket value?

Methodology

- Data Preprocessing
- EDA Analysis
- Association rule Mining
 - Transaction Aggregation
 - Incidence Matrix
 - Frequent Itemset Discovery
 - Association Rule

Project Infrastructure

4,042

UNIQUE ITEMS

23,796

TRANSACTIONS

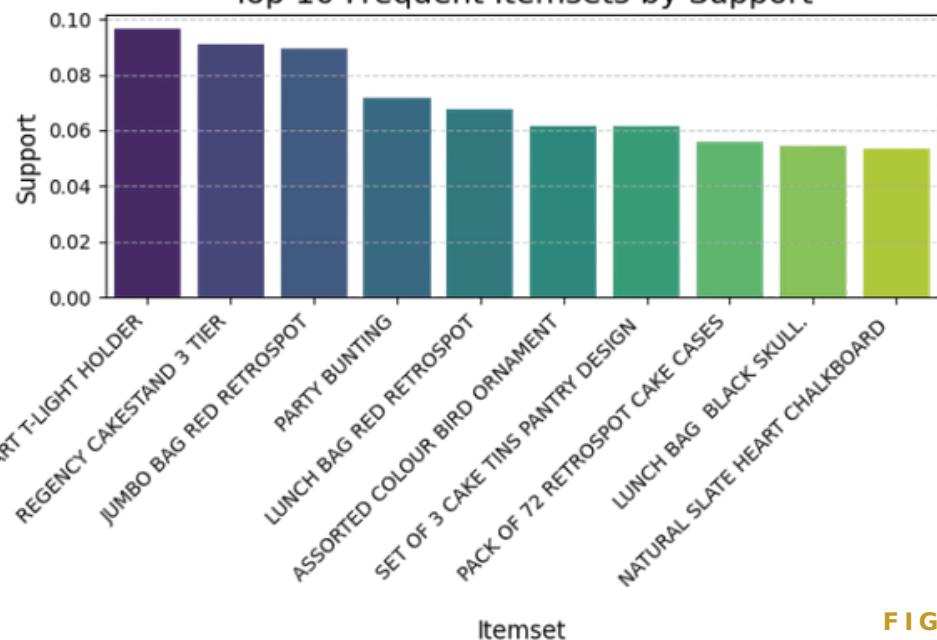
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MIN SUPPORT

A **23,796 × 4,042 sparse matrix** (SciPy CSC) stored only non-zero basket entries for memory-efficient mining at scale.

FIG 01

Top 10 Frequent Itemsets by Support



Key Visualizations & Analysis

Bar chart — Top frequent itemsets

FIG 02

Top 10 Association Rules (sorted by lift):

	antecedents	consequents	support	confidence	lift
36	(GREEN REGENCY TEACUP AND SAUCER, ROSES REGENC...	(PINK REGENCY TEACUP AND SAUCER)	0.023	0.700	20.818
37	(PINK REGENCY TEACUP AND SAUCER)	(GREEN REGENCY TEACUP AND SAUCER, ROSES REGENC...	0.023	0.685	20.818
38	(GREEN REGENCY TEACUP AND SAUCER)	(PINK REGENCY TEACUP AND SAUCER, ROSES REGENC...	0.023	0.519	20.164
35	(PINK REGENCY TEACUP AND SAUCER, ROSES REGENC...	(GREEN REGENCY TEACUP AND SAUCER)	0.023	0.894	20.164
32	(GARDENERS KNEELING PAD KEEP CALM)	(GARDENERS KNEELING PAD CUP OF TEA)	0.023	0.594	18.482
33	(GARDENERS KNEELING PAD CUP OF TEA)	(GARDENERS KNEELING PAD KEEP CALM)	0.023	0.718	18.482
13	(GREEN REGENCY TEACUP AND SAUCER)	(PINK REGENCY TEACUP AND SAUCER)	0.027	0.609	18.129
12	(PINK REGENCY TEACUP AND SAUCER)	(GREEN REGENCY TEACUP AND SAUCER)	0.027	0.804	18.129
39	(ROSES REGENCY TEACUP AND SAUCER)	(PINK REGENCY TEACUP AND SAUCER, GREEN REGENCY...	0.023	0.490	18.124

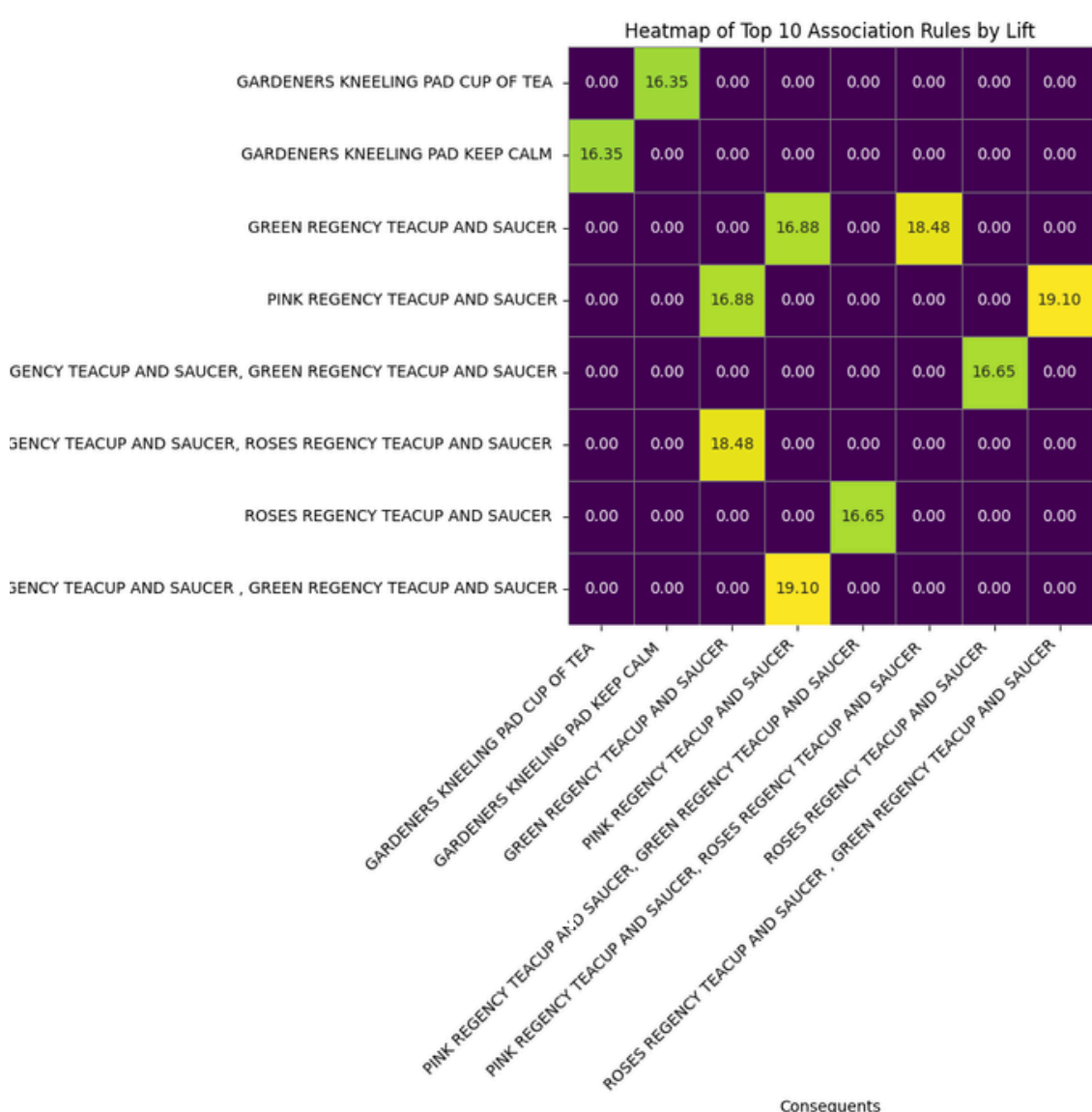
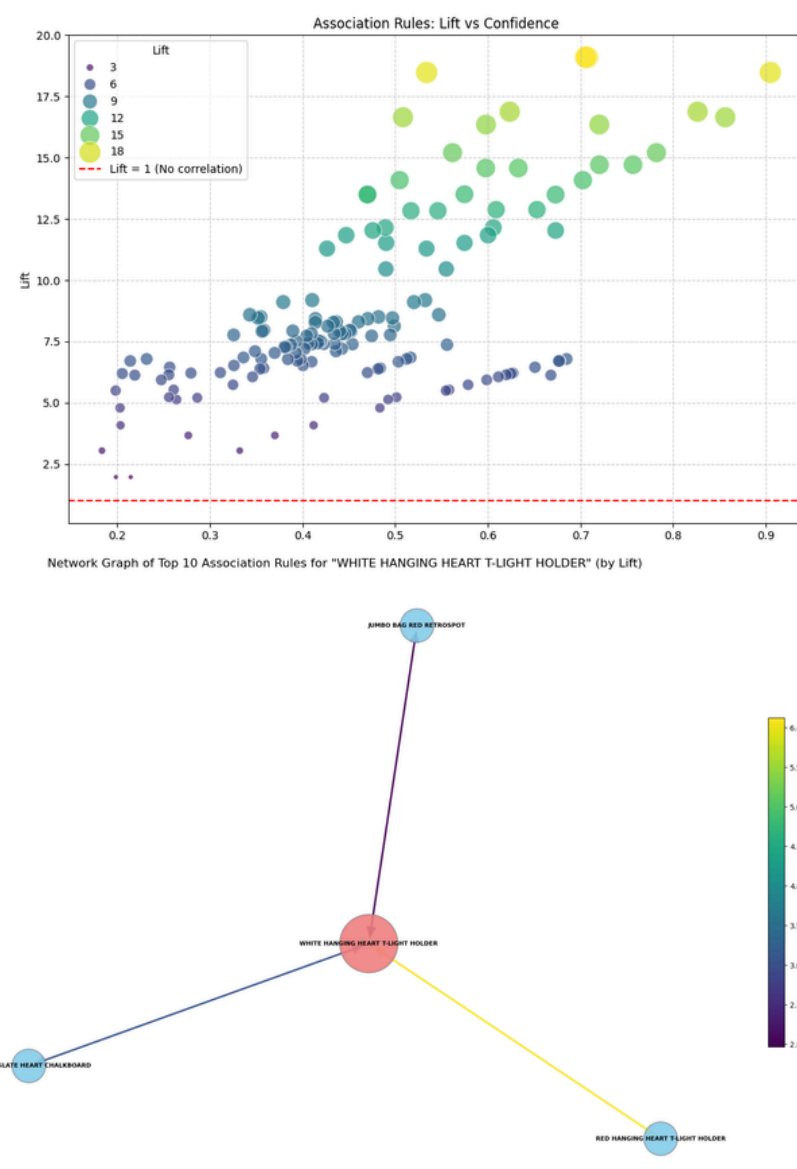


FIG 03



Findings

- High-Frequency Items:** 'WHITE HANGING HEART T-LIGHT HOLDER' has the highest support, indicating strong demand for home décor and gifting products.
- Strong Product Associations:** High lift values (15–19) show that certain items, especially teacup and saucer sets, are frequently purchased together.
- Intentional Buying Behavior:** Moderate to high confidence (0.4–0.9) suggests customers are intentionally bundling items, not purchasing randomly.
- Anchor Product Identified:** The 'WHITE HANGING HEART T-LIGHT HOLDER' acts as a central product, strongly linked to complementary items, making it key for cross-selling.
- Thematic Purchasing Patterns:** Customers tend to buy coordinated, aesthetic, or matching items, particularly within home décor and gift categories.

Limitations

- Lack of Customer Context (Customer demographics)
- No Time Dimension (No analysis of seasonality or trends over time)
- No Pricing or Profitability Insight (analysis does not consider: Product margins and Revenue contribution)
- High Lift ≠ High Sales Volume (Some rules have very high lift but may occur in fewer transactions)

Recommendations

i. Strategic Product Bundling

Package high-lift item pairs, such as a Regency Tea Set or Celebration Set, and promote them at a modest discount to capture already-correlated demand.

ii. Optimised Placement

Position associated items adjacent in-store and surface them as “customers who bought X also bought Y” online to shorten the path to complementary purchases.

iii. Targeted Marketing

Trigger email and on-site recommendations using strong antecedent-consequent rules so personalization is grounded in observed basket behavior.

TAKEAWAY

“Association rule mining turns 23,796 everyday baskets into a concrete playbook for bundling, placement, and personalised promotion.”